

# Week 16: Bonus: categories and functors

Functional Programming in Lean

# Learning goals

- ▶ Identity and composition
- ▶ Category laws
- ▶ Functor laws for `Option`

# Core concepts

- ▶ Objects: types
- ▶ Morphisms: functions
- ▶ Functor mapping: types and morphisms

# Lean example

```
def compose (f :  $\alpha \rightarrow \beta$ )  
  (g :  $\beta \rightarrow \gamma$ ) :  $\alpha \rightarrow \gamma$  :=  
  fun x => g (f x)
```

```
theorem option_map_id (x : Option  $\alpha$ ) :  
  id <$> x = x := by  
  cases x <;> rfl
```

# Lean details

- ▶ Composition associativity
- ▶ `Option.map` preserves identity
- ▶ `Option.map` preserves composition

# Live coding

- ▶ `compose_assoc` with `funext`
- ▶ `option_map_id` by cases
- ▶ `option_map_comp` by cases

# Check

`Option.map` applied to `id`

# Exercise

- ▶ Composition-law proof
- ▶ Tree-map law